Name: 5.2 New Scientific Insights

<Files\\Processed x Highly relevant\\2024-06-20 - RNZ - New AI video tech- It brings a> - § 1 reference coded [2.85% Coverage]

Reference 1 - 2.85% Coverage

Dr Lensen said there were some benefits to the groundbreaking technology, but he thought the cons outweighed the pros.
"Some people talk about things like historical reanimations... potential medical applications around visualising cancers and so on," he explained.

<Files\\Processed x Highly relevant\\2024-09-15 - MBE - How artificial intelligence ma> - § 2 references coded [14.88% Coverage]

Reference 1 - 6.25% Coverage

The project began with photographs taken in nesting boxes at Zealandia, perfectly set up to allow snapshots of the birds heads in profile while they fed.
They were now achieving 90 to 95% accuracy with this method, Lensen said, but in a surprise twist, the AI was using posture rather than beaks, which the team had initially thought would be easiest, to tell birds apart.

Reference 2 - 8.63% Coverage

Three proposals from the Victoria University faculty of engineering had received MBIE funding this year, all using what Lensen called "AI for good": the kÄ kÄ project, another using AI to recognise kiwi calls and another for ice sheet modelling.
"It's a reflection, I think, of where we are as academics," he said. "In contrast to in industry, where there is very much a capitalistic focus, we are quite privileged in that we have this freedom, or in some ways this mandate, to do good things with the technology."

<Files\\Processed x Highly relevant\\2024-10-06 - PRS - AI transforming aged care by p> - § 2 references coded [8.80% Coverage]

Reference 1 - 2.71% Coverage

By analysing more than a million health assessments, an artificial intelligence (AI) programme can now predict injuries, dementia and even mortality for elderly Kiwis.

Reference 2 - 6.09% Coverage

Each hip fracture can result in three weeks in hospital and cost $47,000. In 2014, it was estimated that hip fractures cost New Zealandâ€™s overall health system $171 million.
InterRAI data revealed certain medications such as sedatives were associated with 50% of hip fractures. Alternative treatments could now be considered for patients most of risk of a future fracture.

<Files\\Processed x Highly relevant\\2024-10-08 - STF - Nobel Prize in chemistry award> - § 2 references coded [8.42% Coverage]

Reference 1 - 3.82% Coverage

Hassabis and Jumper created an artificial intelligence model that has been able to predict the structure of virtually all the 200 million proteins that researchers have identified, the committee added.

Reference 2 - 4.60% Coverage

â€œFour years ago in 2020, Demis Hassabis and John Jumper managed to crack the code with skillful use of artificial intelligence. They made it possible to predict the complex structure of essentially any known protein in nature,â€ Linke said.

<Files\\Processed x Highly relevant\\2024-10-30 - RNZ - NIWA turning to AI to forecast> - § 1 reference coded [3.57% Coverage]

Reference 1 - 3.57% Coverage

NIWA is now focusing on whether the model needs to be retrained for each area of the country or if the model can use what it has learnt from Westport to develop forecasts for other areas.

<Files\\Processed x Highly relevant\\2024-11-27 - TAM - Whale song close to being d> - § 2 references coded [2.77% Coverage]

Reference 1 - 0.69% Coverage

Now researchers are on the verge of using arti~cial intelligence to decode the language of cetaceans.

Reference 2 - 2.07% Coverage

He says the big data collected in supercomputers and AI tools such as ChatGPT and Tron have seen the international science community fast-track to decipher the complex vocalisations of whales, dolphins, and other cetaceans. However, Duarte says understanding the language is still a process in progress

<Files\\Processed x Highly relevant\\2025-02-25 - RNZ - New generation AI tool develop> - § 2 references coded [18.28% Coverage]

Reference 1 - 2.64% Coverage

The Whakaari/White Island tragedy has led scientists to develop an eruption predictor tool using AI, which they hope will prevent another fatal volcanic disaster.

Reference 2 - 15.63% Coverage

The seismic tools were the same as those used to measure earthquakes, Ardid said, but near a volcano the recorded shaking typically related to subsurface volcanic activity such as magma movement, which could be a precursor to an eruption.
"When you train these machine learning models you want to use volcanoes that have a lot of eruptions recorded ... because you need to train them with examples of the stuff that you want them to recognise in real time. "In that sense, the New Zealand volcanoes are quite exceptional, especially Whakaari - it's super active, we have great records from [monitoring organisation] Geonet."
Study co-lead associate professor David Dempsey and volcanologist Ben Kennedy said the tool was challenging the idea that eruption warning signs were unique to individual volcanoes, which meant patterns identified at data-rich volcanoes could be used to predict eruptions at volcanoes that were data-poor, such as Mount Taranaki.

<Files\\Processed x Highly relevant\\2025-02-25 - WHC - AI models might help to predic> - § 3 references coded [11.84% Coverage]

Reference 1 - 4.42% Coverage

As with earthquakes, uncertainty and variability have made it essentially impossible for scientists to predict the likelihood and timing of such events. But the specific pre-eruption cues identified by the teamâ€™s machine-learning model may ultimately help change that.

Reference 2 - 5.48% Coverage

Crucially, the warning signals were shown to follow repeatable patterns across multiple volcanoes â€" challenging the current paradigm that each volcano has its own unique patterns.
That meant the model could be applied to less-studied volcanoes elsewhere in the world, where some 29 million people live in areas at risk of eruption.

Reference 3 - 1.93% Coverage

â€œThis finding could be a breakthrough for eruption forecasting,â€ the studyâ€™s lead author, Dr Alberto Ardid, said.

<Files\\Processed x Highly relevant\\2025-02-27 - RNZ - Baby steps for AI volcano fore> - § 2 references coded [3.75% Coverage]

Reference 1 - 2.63% Coverage

A volcanologist says a new artificial intelligence (AI) volcano forecasting tool, with the potential to save lives around the world, is still some way off being rolled out in places like Tonga.

Reference 2 - 1.13% Coverage

It uses data from well-monitored volcanoes and applies it to under-monitored sites.

<Files\\Processed x Highly relevant\\2025-04-26 - RNZ - Researchers uncover link betwe> - § 2 references coded [4.59% Coverage]

Reference 1 - 2.25% Coverage

University of Otago/Ōtākou Whakaihu Waka researcher, Dr Ashleigh Barrett-Young, told Saturday Morning changes to the blood vessels at the back of the eye could indicate someone is at greater risk of developing dementia.

Reference 2 - 2.33% Coverage

"So we found that both the cellular thickness' and the blood vessel measurements were associated with increased risk for developing dementia based on those risk scores, though the blood vessels did show a stronger association."

<Files\\Processed x Highly relevant\\2025-04-30 - NZH - Start-up signs deal with Briti> - § 1 reference coded [3.39% Coverage]

Reference 1 - 3.39% Coverage

Pyper Vision uses AI (artificial intelligence) and deep learning models to detect trends in fog then matches that with data on the ground. â€œOur mission is to bring clear skies to aviation, no matter the weather,â€ Blythe said.